# Continuous Random Variables

**ANSWER KEY**

## Section 1: Probability Density Functions (PDF)

**1** A continuous random variable $X$ has PDF $f(x) = kx(2 - x)$ for $0 \le x \le 2$, and $0$ otherwise. Find the value of $k$.

**a.** $\frac{3}{4}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{8}$

**d.** $1$

Total probability must equal 1: $\int_0^2 k(2x - x^2)dx = k[x^2 - x^3/3]_0^2 = k(4 - 8/3) = k(4/3) = 1 \Rightarrow k = 3/4$.

**2** For the random variable in the previous question ($k = 3/4$), calculate $P(0 \le X \le 1)$.

**a.** $\frac{1}{2}$ ✓

**b.** $\frac{3}{8}$

**c.** $\frac{1}{4}$

**d.** $\frac{5}{8}$

$P(0 \le X \le 1) = \frac{3}{4}\int_0^1 (2x - x^2)dx = \frac{3}{4}[x^2 - x^3/3]_0^1 = \frac{3}{4}(1 - 1/3) = \frac{3}{4}(2/3) = \frac{1}{2}$.

**3** What is the probability that a continuous random variable takes any single exact value, such as $P(X = c)$?

**a.** $0$ ✓

**b.** $1$

**c.** $f(c)$

**d.** $f'(c)$

For any continuous distribution, the probability of taking any exact single point is zero ($P(X = c) = \int_c^c f(x)dx = 0$).

---

## Section 2: Cumulative Distribution Function $F(x)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**4** Given PDF $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, find the cumulative distribution function $F(x)$ for $0 \le x \le 2$.

**a.** $F(x) = \frac{x^3}{8}$ ✓

**b.** $F(x) = \frac{3x^3}{8}$

**c.** $F(x) = \frac{x^2}{4}$

**d.** $F(x) = \frac{x^3}{2}$

$F(x) = \int_0^x f(t)dt = \int_0^x \frac{3}{8}t^2 dt = \left[\frac{t^3}{8}\right]_0^x = \frac{x^3}{8}$.

**5** For the distribution in the previous question, find the median $m$.

**a.** $m = \sqrt[3]{4} \approx 1.587$ ✓

**b.** $m = 1.0$

**c.** $m = \sqrt{2} \approx 1.414$

**d.** $m = 1.5$

Set $F(m) = 0.5 \Rightarrow \frac{m^3}{8} = 0.5 \Rightarrow m^3 = 4 \Rightarrow m = 4^{1/3} \approx 1.587$.

**6** Given $F(x) = \frac{x^2 - 1}{8}$ for $1 \le x \le 3$, find the PDF $f(x)$ on this interval.

**a.** $f(x) = \frac{x}{4}$ ✓

**b.** $f(x) = \frac{x}{8}$

**c.** $f(x) = \frac{x^2}{8}$

**d.** $f(x) = \frac{2x - 1}{8}$

The PDF is the derivative of the CDF: $f(x) = F'(x) = \frac{d}{dx}\left[\frac{x^2 - 1}{8}\right] = \frac{2x}{8} = \frac{x}{4}$.

---

## Section 3: Mean, Variance & Medians

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**7** Find the expected value $E(X)$ for the PDF $f(x) = 2x$ on $[0, 1]$.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{4}$

**d.** $\frac{1}{3}$

$E(X) = \int_0^1 x f(x) dx = \int_0^1 2x^2 dx = [2x^3/3]_0^1 = 2/3.$

**8** For the distribution $f(x) = 2x$ on $[0, 1]$, calculate the variance $\text{Var}(X)$.

**a.** $\frac{1}{18}$ ✓

**b.** $\frac{1}{9}$

**c.** $\frac{1}{12}$

**d.** $\frac{2}{9}$

$E(X^2) = \int_0^1 2x^3 dx = [x^4/2]_0^1 = 1/2$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 1/2 - (2/3)^2 = 1/2 - 4/9 = 1/18$.

**9** Find the mode of the continuous distribution with PDF $f(x) = 12x^2(1 - x)$ for $0 \le x \le 1$.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{3}{4}$

**d.** $\frac{1}{3}$

The mode is where $f(x)$ is maximized: $f'(x) = 24x - 36x^2 = 12x(2 - 3x) = 0 \Rightarrow x = 2/3$.

---

## Section 4: Piecewise Distributions & Examination Standards

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**10** A continuous random variable has PDF $f(x) = cx$ for $0 \le x < 2$ and $f(x) = c(4 - x)$ for $2 \le x \le 4$. Find $c$.

**a.** $\frac{1}{4}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{1}{8}$

**d.** $1$

The PDF forms a symmetric triangle with base 4 and height $2c$. Area $= \frac{1}{2}(4)(2c) = 4c = 1 \Rightarrow c = 1/4$.

**11** For the symmetric triangular distribution in the previous question, what is $E(X)$?

**a.** **2** ✓

**b.** 1

**c.** 3

**d.** 2.5

By symmetry of the PDF about $x = 2$, the expected value must equal the line of symmetry: $E(X) = 2$.

**12** Find the lower quartile $Q_1$ for the triangular distribution in the previous question ($c = 1/4$).

**a.** $\sqrt{2} \approx 1.414$ ✓

**b.** $1.0$

**c.** $1.5$

**d.** $\sqrt{3} \approx 1.732$

For $x \le 2$: $F(x) = \int_0^x \frac{1}{4} t\,dt = \frac{x^2}{8}$. Set $F(Q_1) = 0.25 \Rightarrow \frac{Q_1^2}{8} = \frac{1}{4} \Rightarrow Q_1^2 = 2 \Rightarrow Q_1 = \sqrt{2} \approx 1.414$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647